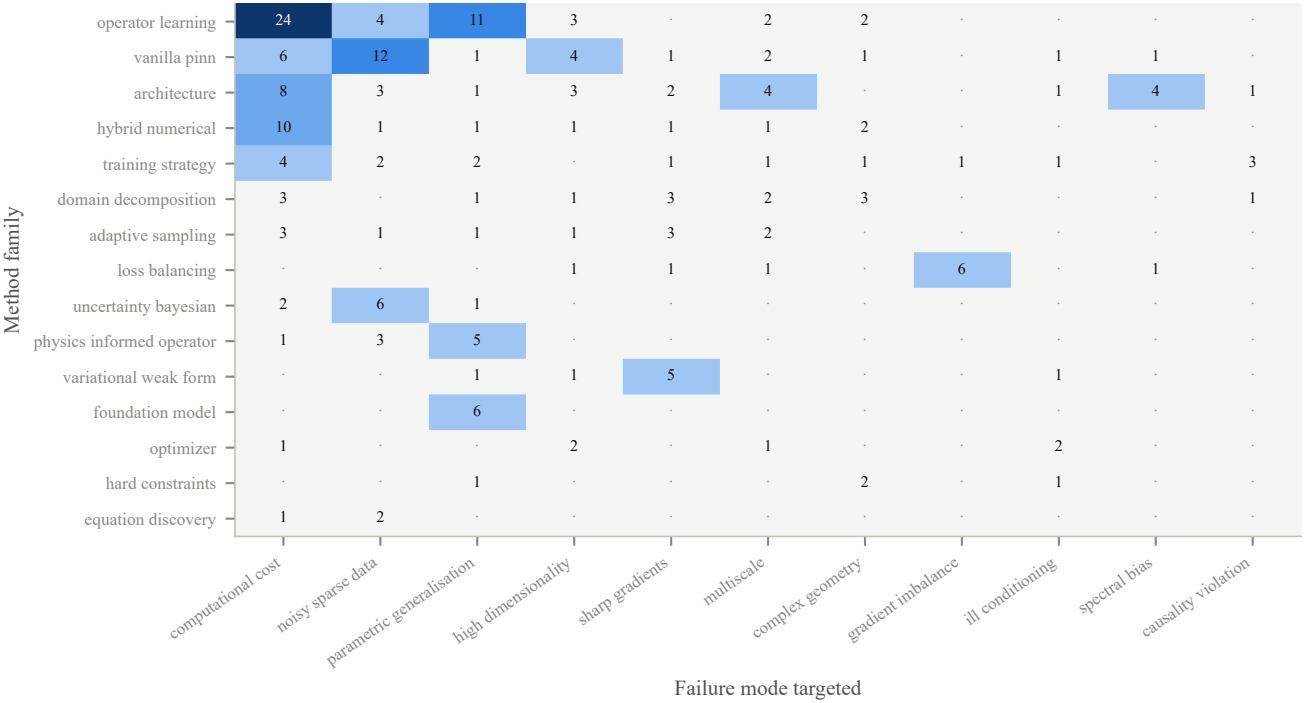
Method family
operator learning
vanilla pinn
architecture
hybrid numerical
training strategy
domain decomposition
adaptive sampling
loss balancing
uncertainty bayesian
physics informed operator
variational weak form
foundation model
optimizer
hard constraints
equation discovery
24 4 11 3 2 2
6 12 1 4 1 2 1 1 1
8 3 1 3 2 4 1 4 1
10 1 1 1 1 1 2
4 2 2 1 1 1 1 1 3
3 1 1 3 2 3 1
3 1 1 1 3 2
1 1 1 6 1
2 6 1
1 3 5
1 1 5 1
6
1 2 1
2 1
1 1 2
computational cost
noisy sparse data
parametric generalisation
high dimensionality
sharp gradients
multiscale
complex geometry
gradient imbalance
ill conditioning
spectral bias
causality violation
Failure mode targeted